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EDUCATION

Bachelor of Science, Data Science | University of California San Diego | GPA: 3.95 – Provost Honors Expected Mar 2027
Relevant Coursework: Applied Data Science, Machine Learning Theory, Recommender Systems, Statistical Inference
Activities and Societies: Hedge Fund Club, Cognitive Science Student Association, Peaks and Professors

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C++, HTML, CSS, JavaScript

Libraries/Frameworks: NumPy, pandas, scikit-learn, PyTorch, TensorFlow, OpenCV, Plotly, LangChain, Jupyter, Git/GitHub

Core Competencies: Data Cleaning/Visualization, EDA, Web Scraping, A/B Testing, Financial Modeling, Time Series Analysis

EXPERIENCE

Machine Learning Researcher

Oct 2025 – Present

Engineers for Exploration

San Diego, CA

- Engineered ensemble machine learning models (70% mIoU baseline) for automated mangrove mapping, developing end-to-end image segmentation pipelines to process satellite and drone imagery for pixel-level land-cover classification.
- Integrated 7 geospatial datasets (SpaceNet, xView, LandCover.ai) to optimize model architecture and feature engineering, accelerating ML-assisted labeling workflows across 4 environmental classes (road, building, water, mangrove).

AI Research & Development Intern

Oct 2025 – Present

Computer Science & Engineering Society

San Diego, CA

- Building AgentSpace, an AI platform leveraging GraphRAG architecture and game-theoretic personality modeling to model professional decision-making patterns, with initial focus on prototyping intelligent agentic tutoring systems.
- Developed retrieval agent prototype using LangChain and RAG pipelines within cross-functional team of 8, experimenting across 6 Gemini model configurations to benchmark performance and optimize response generation workflows.

Research Data Analyst

May 2025 – Present

Scripps Institution of Oceanography

San Diego, CA

- Processed and annotated 9,000+ georeferenced coral imagery points using in-house photogrammetry software (Viscore), maintaining structured datasets for downstream temporal analysis across 15 reef sites on Palmyra Atoll.
- Executed time series analysis across 16 time points (2014-2025) quantifying impact of depth and 5 treatment variables on reef health metrics, delivering insights on coral bleaching trends to inform data-driven restoration strategies.

Quantitative Analyst

Oct 2025 – Dec 2025

Sustainable Investment Group

San Diego, CA

- Implemented automated Python data pipeline using FinBERT NLP, Yahoo Finance API, and Tavily to generate trading signals by quantifying lagged correlations between 25 weeks of news sentiment and 5+ market KPIs (returns, volatility, volume).
- Co-led quantitative investment thesis for Under Armour, integrating ARIMA time series forecasting, sentiment analysis, and DCF financial modeling to deliver data-driven sell recommendation in competitive 5-team stock pitch.

Market Research Intern

Jul 2023 – Aug 2023

Bennett Coleman & Co. Ltd. (The Times of India)

Mumbai, India

- Conducted quantitative research on 56,000+ newspaper headlines using TF-IDF vectorization and multi-output linear regression, training a model to predict relevance scores across 5 key market segments.
- Delivered 5-category classification prototype achieving 85% accuracy, enabling automated headline categorization across Business, Politics, Technology, Social Issues, and Lifestyle segments for editorial workflows.

PROJECTS

NextMove: Predicting Next Workout Activity

Nov 2025 – Dec 2025

Course Project – Recommender Systems

[View Project](#)

- Engineered temporal, behavioral, and physiological features from 20,000+ sequential workout records in the Endomondo dataset, designing transition-based training instances to predict next activity type without look-ahead bias.
- Compared heuristic baselines (frequency, recency, weighted) against supervised ML models (Logistic Regression, Naïve Bayes, SVM, Random Forest), achieving 87% accuracy with Random Forest using chronological splits and balanced class weighting.

ENSOCast: Interactive Climate Analytics Dashboard

May 2025 – Oct 2025

Personal Project (Presented at San Diego Undergraduate Tech Conference 2025)

[View Project](#)

- Built a Streamlit dashboard with interactive Plotly visualizations for exploring 40+ years of ocean temperature data and ENSO phases, enabling users to train and compare custom ML models.
- Developed and evaluated advanced ML models, including Random Forest, XGBoost, 1D CNN, LSTM, and Ensemble Learning, using temporal, seasonal, and interaction-based features, achieving up to 82% accuracy in ENSO phase prediction.